CSA6303 - THREAT INTELLIGENCE AND NETWORK SECUR

Assessment Tool 1: CONCEPT MAPPING

Faculty : Dr.V.Parthipan **SLOT A**

Topics: Network Security Fundamentals | Security Goals (CIA Triad) | Types of Network Threats |Introduction to Threat Intelligence Types of Threat Intelligence (Strategic, Tactical, perational) — Threat Intelligence Lifecycle — Applications in IT Industry

CO1: Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.

Total: 50

Student Name :VELUGU KIRAN BOJA RAJU **Register number:192312353**

Code of Conduct:

I **Velugu Kiran Boja Raju/192312353** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:V.Kiran

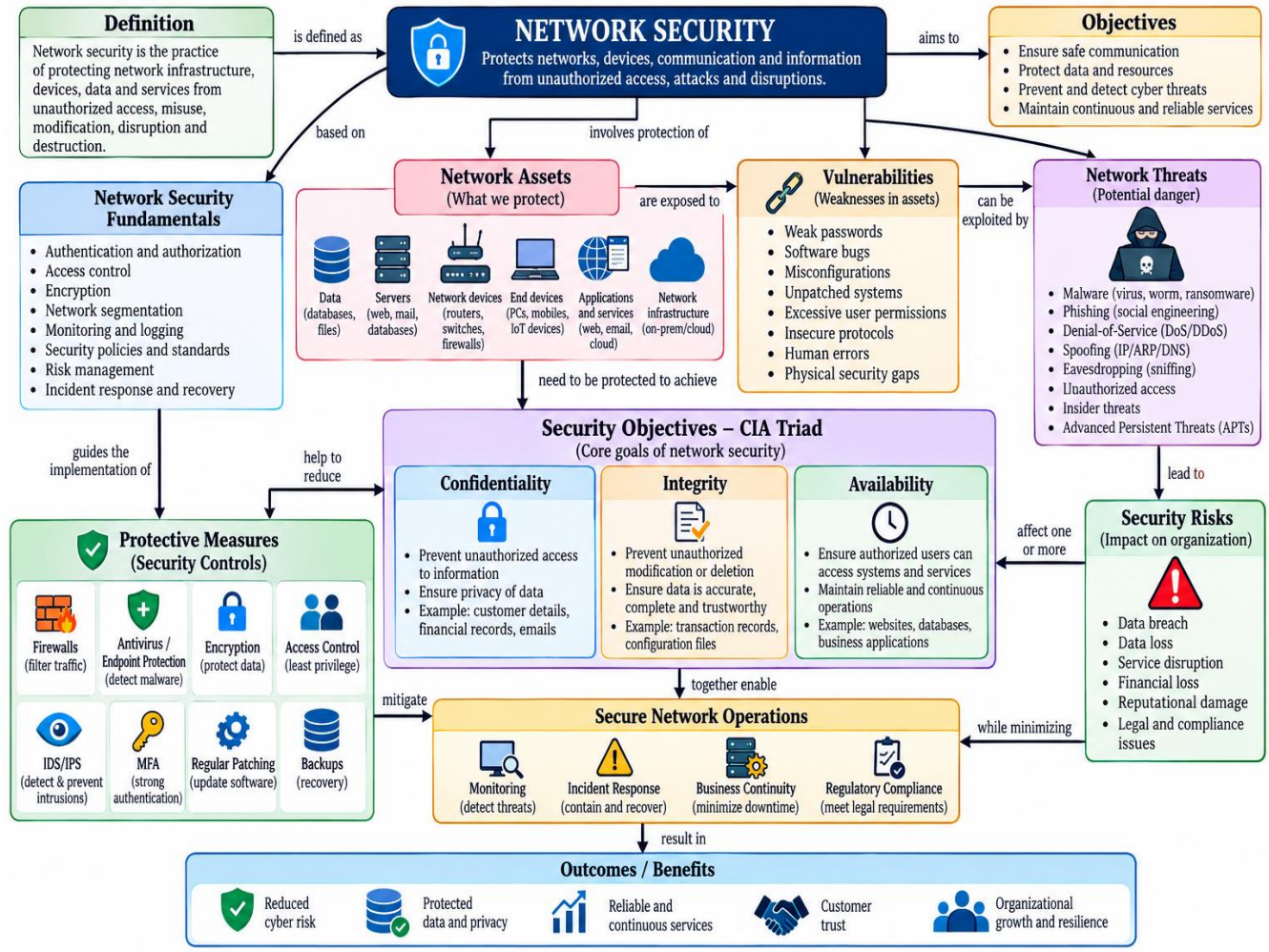
CO1 – Concept Mapping Questions

No.	Concept-Mapping Question
1	**Construct a comprehensive concept map explaining the foundations of Network Security.** Connect network security fundamentals with security objectives, the **CIA Triad (Confidentiality, Integrity, and Availability)**, network assets, vulnerabilities, threats, and protective measures, and explain how these concepts collectively support secure network operations.
2	**Develop a concept map illustrating the CIA Triad as the foundation of network security.** Relate Confidentiality, Integrity, and Availability to different types of network threats and demonstrate how a single threat can affect one or more security goals.
3	**Construct a concept map classifying different types of Network Threats and their potential impact on security goals.** Connect threats such as malware, phishing, denial-of-service, spoofing, eavesdropping, and unauthorized access with the CIA Triad and explain the resulting security implications.
4	**Develop an integrated concept map connecting Network Security Fundamentals, Network Threats, and Threat Intelligence.** Show how identifying assets, vulnerabilities, and threats leads to the collection and analysis of threat information and ultimately supports informed security decisions.
5	**Construct a concept map explaining the concept of Threat Intelligence and its role in modern network security.** Relate threat actors, indicators, vulnerabilities, attack patterns, threat information, analysis, and actionable intelligence to demonstrate how Threat Intelligence helps organizations anticipate and respond to network threats.
6	**Develop a comparative concept map of Strategic, Tactical, and Operational Threat Intelligence.** Connect each type with its purpose, target audience, time horizon, information requirements, decision-making process, and appropriate network-security use cases, and analyze how the three intelligence types complement one another.
7	**Construct a concept map representing the complete Threat Intelligence Lifecycle.** Connect planning/direction, collection, processing, analysis, dissemination, and feedback with the identification of network threats and security requirements, showing how feedback continuously improves the intelligence process.
8	**Develop an end-to-end concept map showing how a network threat progresses from identification to actionable Threat Intelligence.** Connect network security fundamentals, CIA security goals, threat detection, intelligence collection, processing, analysis, dissemination, and response, and explain the relationship between each stage.
9	**Create a scenario-based concept map for an organization experiencing a major network attack.** Integrate the CIA Triad, types of network threats, Strategic Intelligence, Tactical Intelligence, Operational Intelligence, and the Threat Intelligence Lifecycle to demonstrate how the organization can understand, prioritize, and respond to the threat.
10	**Design a comprehensive concept map demonstrating the applications of Network Security and Threat Intelligence in the IT industry.** Connect network security fundamentals, CIA Triad, network threats, Threat Intelligence, Strategic/Tactical/Operational Intelligence, and the Threat Intelligence Lifecycle with

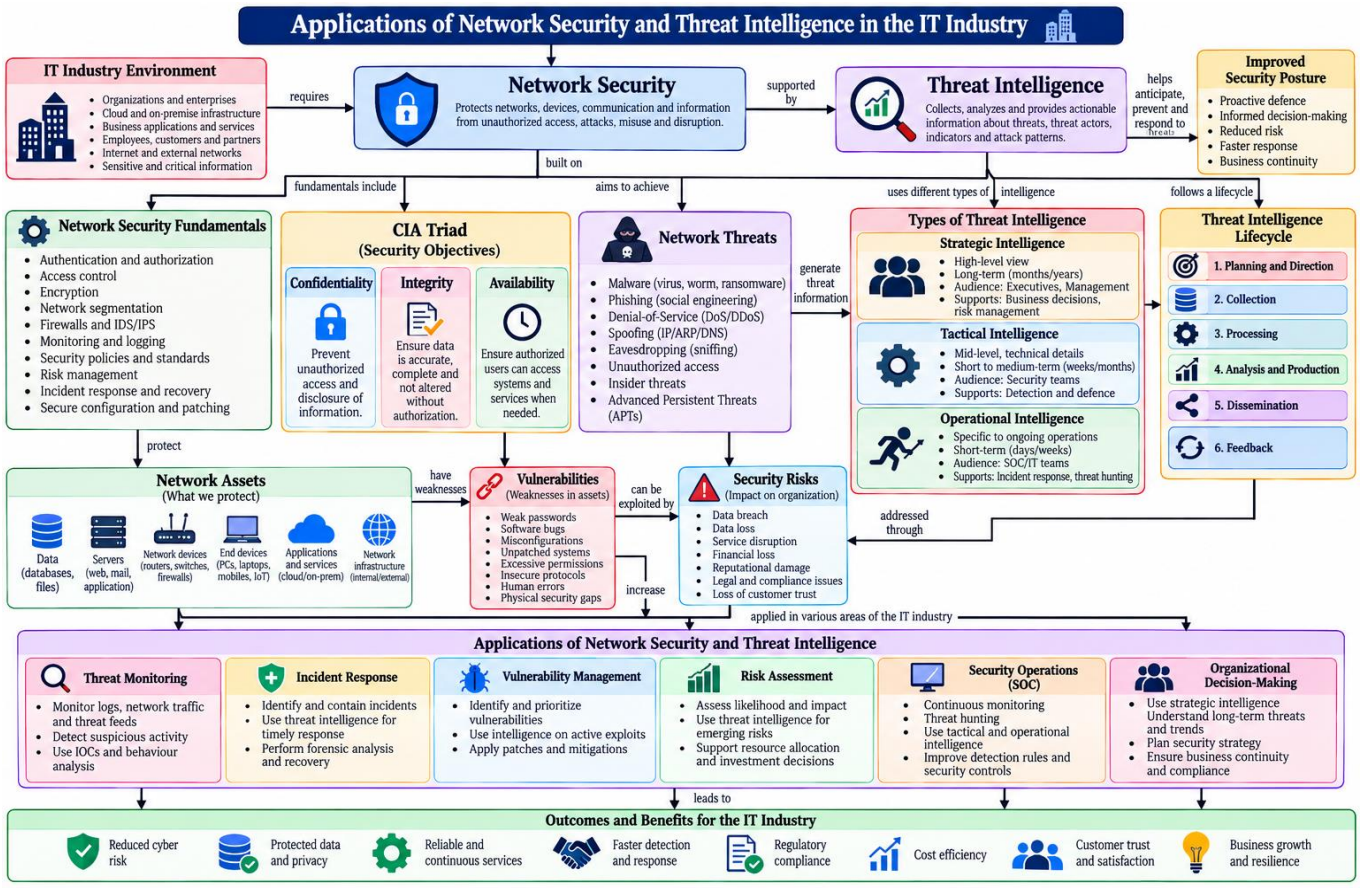
No.	Concept-Mapping Question
	applications such as threat monitoring, incident response, vulnerability management, risk assessment, security operations, and organizational decision-making.

CONCEPT MAP

1A



10A


Applications of Network Security and Threat Intelligence in the IT Industry
IT Industry Environment
Organizations and enterprises
Cloud and on-premise infrastructure
Business applications and services
Employees, customers and partners
Internet and external networks
Sensitive and critical information
requires
Network Security
Protects networks, devices, communication and information from unauthorized access, attacks, misuse and disruption.
supported by
Threat Intelligence
Collects, analyzes and provides actionable information about threats, threat actors, indicators and attack patterns.
helps anticipate, prevent and respond to threats
Improved Security Posture
Proactive defence
Informed decision-making
Reduced risk
Faster response
Business continuity
built on
fundamentals include
aims to achieve
uses different types of intelligence
follows a lifecycle
Network Security Fundamentals
Authentication and authorization
Access control
Encryption
Network segmentation
Firewalls and IDS/IPS
Monitoring and logging
Security policies and standards
Risk management
Incident response and recovery
Secure configuration and patching
CIA Triad
(Security Objectives)
Confidentiality
Prevent unauthorized access and disclosure of information.
Integrity
Ensure data is accurate, complete and not altered without authorization.
Availability
Ensure authorized users can access systems and services when needed.
Network Threats
Malware (virus, worm, ransomware)
Phishing (social engineering)
Denial-of-Service (DoS/DDoS)
Spoofing (IP/ARP/DNS)
Eavesdropping (sniffing)
Unauthorized access
Insider threats
Advanced Persistent Threats (APTs)
generate threat information
Types of Threat Intelligence
Strategic Intelligence
High-level view
Long-term (months/years)
Audience: Executives, Management
Supports: Business decisions, risk management
Tactical Intelligence
Mid-level, technical details
Short to medium-term (weeks/months)
Audience: Security teams
Supports: Detection and defence
Operational Intelligence
Specific to ongoing operations
Short-term (days/weeks)
Audience: SOC/IT teams
Supports: Incident response, threat hunting
Threat Intelligence Lifecycle
1. Planning and Direction
2. Collection
3. Processing
4. Analysis and Production
5. Dissemination
6. Feedback
protect
Network Assets
(What we protect)
Data (databases, files)
Servers (web, mail, application)
Network devices (routers, switches, firewalls)
End devices (PCs, laptops, mobiles, IoT)
Applications and services (cloud/on-prem)
Network infrastructure (internal/external)
have weaknesses
Vulnerabilities
(Weaknesses in assets)
Weak passwords
Software bugs
Misconfigurations
Unpatched systems
Excessive permissions
Insecure protocols
Human errors
Physical security gaps
can be exploited by
Security Risks
(Impact on organization)
Data breach
Data loss
Service disruption
Financial loss
Reputational damage
Legal and compliance issues
Loss of customer trust
increase
addressed through
applied in various areas of the IT industry
Applications of Network Security and Threat Intelligence
Threat Monitoring
Monitor logs, network traffic and threat feeds
Detect suspicious activity
Use IOCs and behaviour analysis
Incident Response
Identify and contain incidents
Use threat intelligence for timely response
Perform forensic analysis and recovery
Vulnerability Management
Identify and prioritize vulnerabilities
Use intelligence on active exploits
Apply patches and mitigations
Risk Assessment
Assess likelihood and impact
Use threat intelligence for emerging risks
Support resource allocation and investment decisions
Security Operations (SOC)
Continuous monitoring
Threat hunting
Use tactical and operational intelligence
Improve detection rules and security controls
Organizational Decision-Making
Use strategic intelligence
Understand long-term threats and trends
Plan security strategy
Ensure business continuity and compliance
leads to
Outcomes and Benefits for the IT Industry
Reduced cyber risk
Protected data and privacy
Reliable and continuous services
Faster detection and response
Regulatory compliance
Cost efficiency
Customer trust and satisfaction
Business growth and resilience